

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 22)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

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0 1 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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0 1 . 3

Explain how auditory monitoring can be used to study bats.

[2 marks]

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0 1 . 4

Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

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0 2 Table 1 describes three genetically modified (GM) crops.

Table 1

GM crop	Gene source	Characteristic
Bt maize / cotton	Bacillus thuringiensis bacterium	Produces a toxin that kills insect pests
Roundup Ready soya	Agrobacterium	Resistant to the herbicide glyphosate
Golden rice	Daffodil and Erwinia bacterium	Produces vitamin A in the grain

0 2 . **1** Explain the main advantage of GM over conventional selective breeding.

[2 marks]

0 2 . **2** For each crop in Table 1, explain one potential benefit.

[3 marks]

Bt crops

Roundup Ready soya

Golden rice

0 2 . **3** Explain **four** concerns about GM crops.

[4 marks]

0 2 . 4 Give **one** reason why GM crops have been slow to enter cultivation in Europe. [1 mark]

10

0 3 Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO ₂ , NO _x

0 3 . 1 Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy. [4 marks]

03 . 2

Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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03 . 3

Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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04 . 1

Describe how agriculture alters the hydrological cycle.

[3 marks]

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0 4 . 2

Explain how each of the following consumer choices can reduce the environmental impact of food production.

[4 marks]

Seasonal / local food

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Organic food

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0 4 . 3

Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

[3 marks]

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0 5

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

05 . 1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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05 . 2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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05 . 3

Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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05 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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06 . 1

Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

- A Close planting of a single non-indigenous species
- B Mixed indigenous species with a mixed age structure
- C Clear-felling on a 40-year rotation
- D Removal of dead wood

☐
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06 . 2

Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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07 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

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07 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

07 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

10

08 . 1

Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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08 . 2

Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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08 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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09 . 1

Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

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09 . 2

Explain **four** reasons why mangroves are important.

[4 marks]

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09 . 3

Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

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Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

Extra space

10 . 2 State what is meant by **rewilding**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Handwriting practice area with 25 horizontal dotted lines.

END OF QUESTIONS